TRUNKBELAY · PRODUCT REQUIREMENTS AND SPECIFICATIONS

TrunkBelay product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 ("i concur with your recommendations"): PR-001 verified by a 3.5 kN proof test and a test to failure of a sample frame (O7); PR-003 lean limit 20 deg and the drill rule for the frame side (O8)
0.4	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: becomes the product requirements and specifications TKB-PRS-001 (replaces TKB-REQ-001); R1 to R10 migrated to PR-001 to PR-010 with targets unchanged, tolerance, units and the user requirement each traces to; PR-011 to PR-018 added for needs the REQ table left implicit; PR-018 records the tether fix TKB-DEC-001 D-S1
0.3	2026-10-03	Amish Chadha	Amish's decisions 16A, 17B, 18C and 19A carried out (TKB-DDR-003); R9 target restated by Amish; status of R2, R4, R7 and R9 updated
0.2	2026-10-03	Amish Chadha	

			Status of every requirement on paper at TRL 3 (TKB-CAL-001); targets unchanged
0.1	2026-09-30	Amish Chadha	Initial scaffold

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What the design must do, each requirement traced to a user need in TKB-URS-001. PR-001 to PR-010 are the requirements R1 to R10 of TKB-REQ-001 with the same targets (R9 as Amish restated it on 2026-10-03); PR-011 to PR-018 make explicit what the design already does. Targets remain proposals for review with a co-design partner. Status comes from the calculation note TKB-CAL-001 v0.3 and the STEM check TKB-STM-001. Method: STANDARDS section 22.

Amish Chadha, 2026-10-03, on the recommendations put to him: "i agree with all the 46 recommendations you provided. please proceed." For TrunkBelay that is decisions 16A (R2: keep the drill rule, double-acting collar as the recorded fallback), 17B (R4: pack the victim set pre-rigged), 18C (R7: the helpers haul the rope bag on a tag line and the collar frame goes to aluminium) and 19A (R9: keep the auto-locking descender and evacuation triangle, seek group purchase, restate the cost target). The design decisions register (TKB-DEC-001) and decision record TKB-DDR-003 hold the detail.

> **Safety:** These requirements describe rope rescue equipment that holds a person at height. Meeting them on paper does not make the kit safe to use. Each is verified by test at TRL 4, and the kit is not used on a person until PR-001 has been proof-tested on cut trunk sections and the drill has been run with a dummy at low height.

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Performance	Collar holds on a palm trunk	The anchor shall hold a 2.5 kN (562 lbf) static downward load on wet trunk sections without slipping more than 20 mm	2.5	Slip 20 mm max	kN	Analysis (TKB-CAL-001 B, C), TRL 3; proof test on cut trunk sections at 3.5 kN and a test to failure of a sample frame, TRL 4 (TKB-DEC-001 O7, decided by Amish 2026-10-25)	Met on paper; the friction on wet bark, settlement (10 mm, 14 to 18 mm with pad shear) and the strength margin (about 5.1:1, 4.5:1 with an unequal chain split; the stronger frame is the fallback if the test to failure shows less than 5:1, O7) are to verify at TRL 4 (locking factor 1.54)	R1
PR-002	UR-002	Safety	Holds when the load direction changes	The anchor shall not slip more than 50 mm when a 100 kg (220 lb) test mass passes the collar during lowering	50	Max	mm	Drop and transition tests on trunk sections, the upward case as a misuse test only, TRL 4	At risk for an upward pull: the single top chain cannot hold it. Decided by Amish (16A): the drill rule stays (collar at least 300 mm above the victim's attachment); the double-acting collar is the recorded fallback	R2
PR-003	UR-003	Physical architecture	Fits common palms	The anchor shall fit trunks 200	200 to 450;	Full range, no tools	mm; deg	Analysis of the model on 200, 300	Met by design for diameter:	R3

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				to 450 mm (8 to 18 in) across, leaning no more than 20 deg from vertical, with no tools; on a leaning trunk the frame is set on the favourable side by the drill rule (lean limit and drill rule decided by Amish, 2026-10-25, TKB-DEC-001 O8)	lean 20 max			and 450 mm trunks (TKB-CAL-001 E2) and of the lock on a leaning trunk (TKB-STM-001 S31), TRL 3; fit trials on surveyed palms and the drill on a leaning palm, TRL 4	the V and the 110-link chain fit the range, 15 links to spare on 450 mm; lean: locking factor 1.21 at 20 deg on the adverse side on the assumed friction, to confirm by test	
PR-004	UR-004	Usability	Quick to rig	A trained climber shall fix the collar and clip the victim in within 5 minutes once in position	5	Max	min	Analysis (TKB-CAL-001 E3), TRL 3; timed drills at 3 m height, TRL 4	At risk: about 4.5 minutes estimated, a thin margin	R4
PR-005	UR-005	Performance	Controlled lowering	The kit shall hold the descent of a 100 kg (220 lb) load under 0.5 m/s with a hand force under 150 N (34 lbf)	150	Max; speed 0.5 m/s max	N	Analysis (TKB-CAL-001 D1), TRL 3; lowering tests with a test mass, TRL 4	Met on paper: about 80 N at the hand with the brake carabiner	R5
PR-006	UR-006	Performance	Rope length	The rope shall lower a person from 25 m (82 ft)	25	Min	m	Inspection and analysis (TKB-CAL-001 E1), TRL 3; measured, TRL 4	Met: 30 m of rope, 28 m needed	R6

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-007	UR-007	Physical architecture	Portable	What the rescuer carries up the trunk shall weigh under 5 kg (11 lb) and hang from the harness, hands-free	5	Max	kg	Analysis (TKB-CAL-001 F2), TRL 3; weighing and climbing trial, TRL 4	Met on paper: 4.67 kg carried; the 3.7 kg bag is hauled up on the tag line (18C); whole kit 8.4 kg	R7
PR-008	UR-008	Safety	No damage to the palm	After a full test lowering no bark cut shall be deeper than 5 mm	5	Max	mm	Analysis (TKB-CAL-001 G1; TKB-STM-001 S19, S20), TRL 3; inspection after trials, TRL 4	Met on paper, at risk on green wood: pads 1.9 MPa mean (about 3.8 MPa at the lower edge) against 9.3 MPa air dry, 4.6 to 5.6 MPa green; bare links mark about 0.3 to 1.3 mm (TKB-STM-001 S19 to S21)	R8
PR-009	UR-009	Cost	Cost	Parts shall cost under USD 850 per kit at single-kit prices, keeping the auto-locking descender and the evacuation triangle; group purchase by the training partner to lower it, one kit per climber group	850	Max	USD	Costed bill of materials (TKB-CAL-001 H2), TRL 3	Met on paper: USD 828 per kit (USD 22 under)	R9

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-010	UR-010	Usability	Teachable	Climbers new to the kit shall complete a supervised rescue drill after a half-day session	0.5	Max	day	Training trial with the partner programme, TRL 4	To verify at TRL 4: cannot be shown on paper	R10
PR-011	UR-011	Safety	Holds if the rescuer lets go	The lowering device shall lock and hold the load when the rescuer releases the handle or the brake strand, and when the handle is pulled too far (anti-panic)	100	Held hands-off, min	kg	Inspection of the bought descender's rating and certificate (BOM line 17; EN 341 class A or EN 12841 type C), TRL 3; dummy drill, TRL 4	Met by design	
PR-012	UR-012	Safety	Head up once freed	The victim set shall hold the person head up once the feet are freed: an evacuation triangle (EN 1498 class B) round the hips and a chest sling on one victim carabiner	1	Person, 100 kg	person	Inspection (BOM lines 16, 19, 20; TKB-STM-001 S22), TRL 3; dummy drill, TRL 4	Met by design; fitting time on a head-down dummy to verify at TRL 4	
PR-013	UR-013	Materials	Survives the climate	No part shall corrode through or lose strength in two years of use and humid storage: bare 6082-T6 frame, stainless fixings in rivet nuts	2	Min	years	Analysis (TKB-STM-001 S25, S26), TRL 3; inspection after the trial season, TRL 4	To verify at TRL 4 (corrosion and ageing are plausible, not shown)	

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				with barrier paste, zinc-plated keeper, soft goods stored in the bag						
PR-014	UR-014	Maintenance	Wear parts replaceable	The pads, sleeve, keeper, lock pin, rope and soft goods shall be inspectable by eye and replaceable with hand tools	0	Special tools	tools	Inspection of the build plan (TKB-BLD-001), TRL 3	Met by design	
PR-015	UR-015	Safety	Recognised rescue gear	The bought rescue gear shall meet EN 341 class A or EN 12841 type C (descender), EN 362 class B (carabiners), EN 1891 type A (rope) and EN 1498 class B (triangle) or equal; the kit shall be marked as not certified equipment	4	All four items	items	Inspection of the BOM and the marking on the spine (TKB-BLD-001 sheet TKB-DWG-105), TRL 3	Met by design	
PR-016	UR-016	Physical architecture	Made locally	The made parts shall be cut from plate, sheet, tube and rubber sheet, drilled and AC TIG welded, with no casting,	0	Special processes	processes	Inspection of the build plan (TKB-BLD-001), TRL 3	Met by design (an AC TIG welder is needed)	

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				machining or special tooling						
PR-017	UR-017	Physical architecture	Patent design-around	The collar shall have no ratcheting cinch, cam or cleated gripping plate (3M DBI-SALA Cynch-Lok geometry)	0	Such features	features	Inspection of the drawings and the precis (TKB-PRC-001), TRL 3	Met by design	
PR-018	UR-004	Safety	Fixed end captive	The chain's fixed end shall stay attached to the frame when the keeper is off, without crowding the brake carabiner in the spare eye	1	Tether, min	tether	Inspection of the model (cad/fidelity.json, no interference) and the build plan step 5, TRL 3	Met by design: grade 80 connecting link through the 10 mm tether hole (TKB-DEC-001 D-S1)	

Categories: physical architecture, performance, materials, usability, safety, maintenance, cost. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- A second climber is often on site or nearby when a climber is stranded, as in the reported cases.
- The rescuer can reach and work just below the victim using their own climbing device.
- Coconut bark and fibre can carry a collar load without crushing, if the load is spread (TKB-STM-001 S19).
- Victims can be fitted with a sling while hanging head down.